

# Vectors

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## Vectors

**Exercise 0.1** (Store a data point as one object). A house has a floor area of 120 square meters, 3 bedrooms, and an age of 15 years.

1. Write these three numbers as a single vertical list  $\tilde{x}$ , in the order given, and call its entries  $x_1$ ,  $x_2$  and  $x_3$ . What is  $x_2$ ?
2. Does the list  $(3, 120, 15)$  describe the same house?

### Solution

*Solution 0.1.*

1. Stacking the three numbers in order gives

$$\tilde{x} = \begin{bmatrix} 120 \\ 3 \\ 15 \end{bmatrix},$$

so  $x_2 = 3$ , the number of bedrooms.

2. No. The list  $(3, 120, 15)$  would describe a house with 3 square meters of floor area and 120 bedrooms. An entry's position says what it measures, so the order of the entries matters.

**Definition 0.1** (Column vector). A **column vector** of length  $p$  is an ordered list of  $p$  numbers, written vertically:

$$\tilde{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$$

*Remark 0.1* (Vectors are columns by default). In these notes, as in most statistics textbooks, a vector is a column vector unless stated otherwise. A vector written inline as a list in parentheses is still a column vector: for example,  $\tilde{x} = (2, -1, 5)$  means

$$\tilde{x} = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}.$$

**Definition 0.2** (The set  $\mathbb{R}^p$ ). For a positive integer  $p$ ,  $\mathbb{R}^p$  is the set of all column vectors of length  $p$  whose entries are real numbers.

*Remark 0.2* (Reading  $\tilde{x} \in \mathbb{R}^p$ ). Writing  $\tilde{x} \in \mathbb{R}^p$  says, in one symbol, that  $\tilde{x}$  is a column vector with  $p$  real entries. For example,  $(1, -2, 0.5) \in \mathbb{R}^3$ . The vector  $(1, 2)$  is not in  $\mathbb{R}^3$ , because it has only 2 entries; it is in  $\mathbb{R}^2$ . In Exercise 0.1,  $\tilde{x} \in \mathbb{R}^3$ .

**Definition 0.3** (Scalar). In linear algebra, a **scalar** is a single real number, as opposed to a vector or a matrix, which holds several numbers.

**Example 0.1** (Scalars and a vector). The numbers 5 and  $-0.2$  are scalars. The column vector  $(5, -0.2)$  is not a scalar: it has 2 entries, so it is in  $\mathbb{R}^2$  (Definition 0.2), and each of its 2 entries is a scalar.

**Definition 0.4** (Real-valued (scalar-valued) and vector-valued functions). A function  $f : A \rightarrow \mathbb{R}$ , whose values are numbers, is **real-valued** (also called **scalar-valued**, since its values are scalars). A function  $f : A \rightarrow \mathbb{R}^m$ , whose values are vectors (Definition 0.2), is **vector-valued**.

**Example 0.2** (A real-valued and a vector-valued function). On  $\mathbb{R}^2$ ,  $f(\tilde{x}) = x_1 + x_2$  is real-valued and  $g(\tilde{x}) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}$  is vector-valued:

$$f\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = 3 + 4 = 7, \quad g\left(\begin{bmatrix} 3 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 3 + 4 \\ 3 - 4 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}$$

**Definition 0.5** (Transpose). The **transpose** of a column vector  $\tilde{x}$  is the row vector with the same [sequence](#) of entries, written horizontally:

$$\tilde{x}^\top \equiv \tilde{x}' \equiv [x_1, x_2, \dots, x_p]$$

**Example 0.3** (Transposing a column vector). The transpose of the column vector with entries 2, −1, 5 is the row vector with the same entries:

$$\begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}^\top = [2, -1, 5].$$

**Definition 0.6** (Vector addition). The **sum** of two column vectors  $\tilde{x}$  and  $\tilde{y}$  of the same length  $p$  is the column vector  $\tilde{x} + \tilde{y}$  of length  $p$  obtained by adding entry by entry:

$$(\tilde{x} + \tilde{y})_i \stackrel{\text{def}}{=} x_i + y_i, \quad i = 1, \dots, p$$

**Example 0.4** (Adding two vectors).

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 + 4 \\ 2 + 5 \\ 3 + 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$$

**Definition 0.7** (Dot product). For any two real-valued vectors  $\tilde{x} = (x_1, \dots, x_p)$  and  $\tilde{y} = (y_1, \dots, y_p)$  of the same length  $p$ , the **dot product** of  $\tilde{x}$  and  $\tilde{y}$  is:

$$\tilde{x} \cdot \tilde{y} = \tilde{x}^\top \tilde{y} \stackrel{\text{def}}{=} \sum_{i=1}^p x_i y_i \tag{1}$$

See also the definitions in:

- Dobson and Barnett (2018), Section 1.3 (equation 1.1, page 7)

- Kaplan (2022), chapter on vectors
- [wikipedia](#)

The dot product has a different generalization for two matrices; see [wikipedia](#) for more.

**Example 0.5** (A dot product). For  $\tilde{x} = (1, 2, 3)$  and  $\tilde{y} = (4, 5, 6)$ :

$$\begin{aligned}\tilde{x} \cdot \tilde{y} &= 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 && \text{(definition of the dot product)} \\ &= 4 + 10 + 18 && \text{(multiply)} \\ &= 32 && \text{(add)}\end{aligned}$$

**Definition 0.8** (Linear combination). A **linear combination** of the numbers  $a_1, \dots, a_k$  with **coefficients**  $c_1, \dots, c_k$  is the weighted sum

$$c_1 a_1 + \dots + c_k a_k = \sum_{i=1}^k c_i a_i$$

A linear combination of vectors  $\tilde{v}_1, \dots, \tilde{v}_k$  of the same length is defined entry by entry: entry  $j$  of  $\sum_{i=1}^k c_i \tilde{v}_i$  is the linear combination of the  $j$ th entries of  $\tilde{v}_1, \dots, \tilde{v}_k$  with the same coefficients. For example, the linear combination of  $(1, 0)^\top$  and  $(0, 1)^\top$  with coefficients 2 and 3 is  $(2 \cdot 1 + 3 \cdot 0, 2 \cdot 0 + 3 \cdot 1)^\top = (2, 3)^\top$ .

*Remark 0.3* (The dot product is a linear combination). The dot product  $\tilde{x} \cdot \tilde{y}$  is a linear combination (Definition 0.8) of the entries of  $\tilde{y}$ , with coefficients  $x_1, \dots, x_p$ . In Example 0.5,  $\tilde{x} \cdot \tilde{y} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6$  is the linear combination of 4, 5, and 6 with coefficients 1, 2, and 3.

The dot product is also the standard *inner product* on  $\mathbb{R}^p$ ; “inner product” is the general notion (Definition 1 in Inner Products and Orthogonality, later in these notes), of which the dot product is one example.

**Theorem 0.1** (Dot product is symmetric). *The dot product is symmetric:*

$$\tilde{x} \cdot \tilde{y} = \tilde{y} \cdot \tilde{x}$$

**i** Proof

*Proof.*

$$\begin{aligned}\tilde{x} \cdot \tilde{y} &= \sum_{i=1}^p x_i y_i && \text{(definition of the dot product)} \\ &= \sum_{i=1}^p y_i x_i && \text{(products of numbers are symmetric)} \\ &= \tilde{y} \cdot \tilde{x} && \text{(definition of the dot product)}\end{aligned}$$

□

## Special vectors

**Definition 0.9** (Zero vector). The **zero vector**  $\tilde{0}$  of length  $p$  has all entries equal to zero:

$$\tilde{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

*Remark 0.4* (The zero vector is the additive identity). Adding the zero vector to a vector leaves it unchanged (Definition 0.6):  $\tilde{x} + \tilde{0} = \tilde{x}$  for any vector  $\tilde{x}$  of the same length, so  $\tilde{0}$  is the **identity element** for vector addition. For example,

$$\begin{aligned}(2, -1) + (0, 0) &= (2 + 0, -1 + 0) \\ &= (2, -1).\end{aligned}$$

**Definition 0.10** (Mean (average, sample mean)). The **mean** (or **average**) of  $n$  numbers  $y_1, \dots, y_n$  is their **sum** divided by how many there are:

$$\bar{y} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n y_i$$

When the numbers are observed data values,  $\bar{y}$  is also called the **sample mean**.

**Example 0.6** (The mean of three numbers). For  $y_1 = 2$ ,  $y_2 = -1$  and  $y_3 = 5$ :

$$\begin{aligned}
\bar{y} &= \frac{1}{3} (2 + (-1) + 5) && \text{(definition of the mean, with } n = 3) \\
&= \frac{1}{3} \cdot 6 && \text{(add)} \\
&= 2 && \text{(multiply)}
\end{aligned}$$

**Definition 0.11** (Ones vector). The **ones vector**  $\tilde{\mathbf{1}}$  of length  $p$  has all entries equal to one:

$$\tilde{\mathbf{1}} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

*Remark 0.5* (The ones vector sums the entries). The dot product (Definition 0.7) of  $\tilde{\mathbf{1}}$  with  $\tilde{x}$  is the sum of the entries of  $\tilde{x}$ :

$$\begin{aligned}
\tilde{\mathbf{1}} \cdot \tilde{x} &= \sum_{i=1}^p 1 \cdot x_i \\
&= \sum_{i=1}^p x_i.
\end{aligned}$$

For example, for  $\tilde{x} = (2, -1, 5)$ ,

$$\begin{aligned}
\tilde{\mathbf{1}} \cdot \tilde{x} &= 2 + (-1) + 5 \\
&= 6,
\end{aligned}$$

and dividing by the length  $p = 3$  gives the **mean** of the entries,  $6/3 = 2$ .

**Definition 0.12** (Indicator vector (standard basis vector)). The  $j$ -th **indicator vector** (or **standard basis vector**)  $\tilde{e}_j$  of length  $p$  has a 1 in position  $j$  and 0s elsewhere:

$$(\tilde{e}_j)_i = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \tilde{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow \text{position } j$$

**Remark 0.6** (Indicator vectors build every vector). For  $p = 3$ , the indicator vectors are  $\tilde{e}_1 = (1, 0, 0)$ ,  $\tilde{e}_2 = (0, 1, 0)$ , and  $\tilde{e}_3 = (0, 0, 1)$ . Every vector of length 3 is a weighted sum  $c_1\tilde{e}_1 + c_2\tilde{e}_2 + c_3\tilde{e}_3$  of them, with weights equal to its entries; for example,  $(2, -1, 5) = 2\tilde{e}_1 - 1\tilde{e}_2 + 5\tilde{e}_3$ . No other weights work: entry  $j$  of  $c_1\tilde{e}_1 + c_2\tilde{e}_2 + c_3\tilde{e}_3$  is  $c_j$ , so each weight  $c_j$  must equal entry  $j$  of the vector.

**Theorem 0.2** (Indicator vectors select entries). *For any vector  $\tilde{x}$  of length  $p$  and any  $j \in \{1, \dots, p\}$ :*

$$\tilde{e}_j \cdot \tilde{x} = x_j$$

### **i** Proof

*Proof.* Writing the dot product componentwise:

$$\begin{aligned} \tilde{e}_j \cdot \tilde{x} &= \sum_{i=1}^p (\tilde{e}_j)_i x_i && \text{(definition of the dot product)} \\ &= \sum_{i=1}^p \begin{cases} 1 \cdot x_i & \text{if } i = j \\ 0 \cdot x_i & \text{if } i \neq j \end{cases} && \text{(definition of } \tilde{e}_j) \\ &= x_j && \text{(only the } i = j \text{ term is nonzero)} \end{aligned}$$

□

**Theorem 0.3** (A vector is a linear combination of indicator vectors). *Every vector  $\tilde{x} \in \mathbb{R}^p$  is the linear combination (Definition 0.8) of the indicator vectors  $\tilde{e}_1, \dots, \tilde{e}_p$  (Definition 0.12) with coefficients  $x_1, \dots, x_p$ :*

$$\tilde{x} = \sum_{j=1}^p x_j \tilde{e}_j$$

### **i** Proof

*Proof.* Compare the two sides entry by entry. For each  $i \in \{1, \dots, p\}$ :

$$\begin{aligned}
\left(\sum_{j=1}^p x_j \tilde{e}_j\right)_i &= \sum_{j=1}^p x_j (\tilde{e}_j)_i && \text{(Definition 0.8, entry by entry)} \\
&= \sum_{j=1}^p \begin{cases} x_j \cdot 1 & \text{if } j = i \\ x_j \cdot 0 & \text{if } j \neq i \end{cases} && \text{(Definition 0.12)} \\
&= x_i \cdot 1 + \sum_{j \neq i} x_j \cdot 0 && \text{(split off the } j = i \text{ term)} \\
&= x_i + 0 && \text{(times 1 changes nothing; times 0 gives 0)} \\
&= x_i && \text{(adding 0 changes nothing)}
\end{aligned}$$

□

**Example 0.7** (Expanding a vector in indicator vectors).

$$\begin{aligned}
\begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix} &= 4 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\
&= 4\tilde{e}_1 - \tilde{e}_2 + 2\tilde{e}_3
\end{aligned}$$

## Orthogonality

**Definition 0.13** (Orthogonal vectors). Two vectors  $\tilde{x}$  and  $\tilde{y}$  of the same length are **orthogonal** (written  $\tilde{x} \perp \tilde{y}$ ) if their dot product (Definition 0.7) is zero:

$$\tilde{x} \perp \tilde{y} \iff \tilde{x} \cdot \tilde{y} = 0$$

*Remark 0.7* (Orthogonal means perpendicular). Orthogonality extends the geometric idea of perpendicular lines to vectors with any number of entries. In the plane,  $(1, 2)$  and  $(-2, 1)$  are perpendicular, and their dot product is  $1 \cdot (-2) + 2 \cdot 1 = 0$ . The same test works in  $\mathbb{R}^3$ , where pictures are harder to draw:

$$\begin{aligned}
(1, 1, 0) \cdot (1, -1, 5) &= 1 - 1 + 0 \\
&= 0,
\end{aligned}$$

so  $(1, 1, 0) \perp (1, -1, 5)$ .

**Example 0.8** (Vectors that are not orthogonal).

$$\begin{aligned}(1, 2) \cdot (1, 1) &= 1 \cdot 1 + 2 \cdot 1 \\ &= 3 \\ &\neq 0,\end{aligned}$$

so  $(1, 2)$  and  $(1, 1)$  are not orthogonal.

**Definition 0.14** (Euclidean norm (length,  $L_2$  norm)). The **Euclidean norm** (or **length**, also called the  $L_2$  **norm** and written  $\|\tilde{x}\|_2$ ) of a vector  $\tilde{x}$  of length  $p$  is

$$\|\tilde{x}\| \stackrel{\text{def}}{=} \sqrt{\tilde{x} \cdot \tilde{x}} = \sqrt{\sum_{i=1}^p x_i^2} \quad (2)$$

where  $\sqrt{\cdot}$  is the [square root](#).

**Example 0.9** (The length of a vector). For  $\tilde{x} = (3, 4)$ :

$$\begin{aligned}\|\tilde{x}\| &= \sqrt{3^2 + 4^2} \quad (\text{definition of the norm}) \\ &= \sqrt{25} \quad (\text{square and add}) \\ &= 5\end{aligned}$$

The vector  $(0.6, 0.8)$  has norm  $\sqrt{0.36 + 0.64} = 1$ .

**Definition 0.15** (Unit vector). A **unit vector** is a vector whose Euclidean norm (Definition [0.14](#)) is 1. A unit vector is said to have *unit length*.

**Example 0.10** (A unit vector and a vector that is not one). For  $\tilde{x} = (0.6, 0.8)$ :

$$\begin{aligned}\|\tilde{x}\| &= \sqrt{0.6^2 + 0.8^2} \quad (\text{definition of the norm}) \\ &= \sqrt{0.36 + 0.64} \quad (\text{square}) \\ &= \sqrt{1} \quad (\text{add}) \\ &= 1 \quad (\text{take the square root})\end{aligned}$$

so  $(0.6, 0.8)$  is a unit vector. The vector  $(1, 1)$  is not:

$$\begin{aligned}\|(1, 1)\| &= \sqrt{1^2 + 1^2} \\ &= \sqrt{2} \\ &\neq 1.\end{aligned}$$

**Exercise 0.2** (How far apart are two points?). Let  $\tilde{x} = (1, 2)$  and  $\tilde{y} = (4, 6)$ .

1. Compute the vector  $\tilde{x} - \tilde{y}$ .
2. Compute the length of  $\tilde{x} - \tilde{y}$ , using the [Euclidean norm](#).
3. Is the result the same if you compute the length of  $\tilde{y} - \tilde{x}$  instead?

Solution

*Solution 0.2.*

1.

$$\begin{aligned}\tilde{x} - \tilde{y} &= (1 - 4, 2 - 6) \\ &= (-3, -4).\end{aligned}$$

2.

$$\begin{aligned}\|\tilde{x} - \tilde{y}\| &= \sqrt{(-3)^2 + (-4)^2} \\ &= \sqrt{25} \\ &= 5.\end{aligned}$$

3.  $\tilde{y} - \tilde{x} = (3, 4)$ , and

$$\begin{aligned}\|\tilde{y} - \tilde{x}\| &= \sqrt{3^2 + 4^2} \\ &= 5.\end{aligned}$$

Yes, the result is the same.

**Definition 0.16** (Euclidean distance). The **Euclidean distance** between two vectors  $\tilde{x}$  and  $\tilde{y}$  of length  $p$  is the [Euclidean norm](#) of their difference:

$$d(\tilde{x}, \tilde{y}) \stackrel{\text{def}}{=} \|\tilde{x} - \tilde{y}\| = \sqrt{\sum_{i=1}^p (x_i - y_i)^2} \quad (3)$$

**Remark 0.8** (Distance from the origin). The distance from the origin  $\tilde{0}$  to a vector  $\tilde{x}$  is the norm of  $\tilde{x}$ :

$$\begin{aligned} d(\tilde{x}, \tilde{0}) &= \|\tilde{x} - \tilde{0}\| \\ &= \|\tilde{x}\| \end{aligned}$$

(Goodfellow et al. 2016, 39).

For example, the distance from  $\tilde{0}$  to  $(3, 4)$  is  $\|(3, 4)\| = 5$ , as in Example 0.9.

**Definition 0.17** (Orthonormal vectors). A set of vectors  $\{\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_k\}$  is **orthonormal** if the vectors are mutually orthogonal (Definition 0.13) and each has norm 1 (Definition 0.14), that is, **unit length**:

$$\tilde{x}_i \cdot \tilde{x}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

**Example 0.11** (Indicator vectors are orthonormal). The indicator vectors  $\tilde{e}_1, \tilde{e}_2, \dots, \tilde{e}_p$  (Definition 0.12) form an orthonormal set. By Theorem 0.2,  $\tilde{e}_i \cdot \tilde{e}_j$  is entry  $i$  of  $\tilde{e}_j$ , which is 1 if  $i = j$  and 0 if  $i \neq j$ . For  $p = 2$ :

$$\begin{aligned} \tilde{e}_1 \cdot \tilde{e}_1 &= 1 \cdot 1 + 0 \cdot 0 \\ &= 1, \end{aligned}$$

$$\begin{aligned} \tilde{e}_2 \cdot \tilde{e}_2 &= 0 \cdot 0 + 1 \cdot 1 \\ &= 1, \end{aligned}$$

and

$$\begin{aligned} \tilde{e}_1 \cdot \tilde{e}_2 &= 1 \cdot 0 + 0 \cdot 1 \\ &= 0. \end{aligned}$$

**Example 0.12** (Two ways to fail to be orthonormal).

- $(1, 1)$  and  $(1, -1)$  are orthogonal, since  $1 \cdot 1 + 1 \cdot (-1) = 0$ , but not orthonormal:  $(1, 1) \cdot (1, 1) = 2 \neq 1$ .
- $(1, 0)$  and  $(0.6, 0.8)$  both have unit length, since  $(1, 0) \cdot (1, 0) = 1$  and

$$\begin{aligned} (0.6, 0.8) \cdot (0.6, 0.8) &= 0.36 + 0.64 \\ &= 1, \end{aligned}$$

but they are not orthonormal:  $(1, 0) \cdot (0.6, 0.8) = 0.6 \neq 0$ .

**Exercise 0.3** (Compute an inner product and a norm). Let

$$\tilde{a} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \quad \tilde{b} = \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix}$$

Compute  $\tilde{a}^\top \tilde{b}$  and  $\|\tilde{a}\|_2$ .

Solution

*Solution 0.3.* Multiply entry by entry and add:

$$\begin{aligned} \tilde{a} \cdot \tilde{b} &= (3)(0) + (-1)(4) + (2)(-2) \\ &= 0 - 4 - 4 \\ &= -8 \end{aligned}$$

For the norm, take the inner product of  $\tilde{a}$  with itself first:

$$\begin{aligned} \tilde{a} \cdot \tilde{a} &= 3^2 + (-1)^2 + 2^2 \\ &= 9 + 1 + 4 \\ &= 14 \end{aligned}$$

so

$$\|\tilde{a}\|_2 = \sqrt{14} \approx 3.742$$

The inner product came out negative while the norm cannot: a norm is a square root of a sum of squares, so it is never below zero.

Python has both operations built in, and R builds them from `sum`.

```
import numpy as np

a = np.array([3.0, -1.0, 2.0])
b = np.array([0.0, 4.0, -2.0])

print(a @ b, round(float(np.linalg.norm(a)), 3))
```

```
a <- c(3, -1, 2)
b <- c(0, 4, -2)

print(c(sum(a * b), round(sqrt(sum(a^2)), 3)))
#> [1] -8.000 3.742
```

Both print  $-8$  and  $3.742$ .

#### Video lecture

Hutchinson's [Linear Function Basics](#) (24 min) covers dot products and the Euclidean norm, and goes on to hyperplanes ([Definition 19 in Matrices](#)) and [level sets](#) (Hutchinson, n.d.). The login for the video site is posted [on Canvas](#).

Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. <https://doi.org/10.1201/9781315182780>.

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